

## Singular Value Decomposition (SVD)

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

Singular Valued Decomposition, SVD, general linear maps

## Singular Value Decomposition

- One of the most important results in linear algebra is that any matrix  $X$  can be decomposed as

$$X = USV^T$$

- If  $X$  is an  $m \times p$  matrix then  $U$  is an  $m \times m$  orthogonal matrix and  $V$  is a  $p \times p$  orthogonal matrix i.e.  $UU^T = U^T U = I$ ,  $VV^T = V^T V = I$
- And  $S$  is an  $m \times p$  matrix with non-zero diagonal elements  $S_{ii} = s_i$  known as the **singular values**
- Where did that come from?

## Eigenvectors

- Note that as  $Xv = su$  and  $X^T u = sv$  then

$$X(-v) = (-s)u \quad X^T u = (-s)(-v)$$

if  $\begin{pmatrix} u \\ v \end{pmatrix}$  is an eigenvector of  $B$  with eigenvalue  $s$  then so is  $\begin{pmatrix} u \\ -v \end{pmatrix}$  with eigenvalue  $-s$

- If  $m < p$  then  $X^T X$  is not full rank so some eigenvalues are zero
- As a consequence  $p - m$  vectors exist such that  $Xv = 0$
- The eigenvalues and eigenvectors are

$$m \times \left( s_i, \begin{pmatrix} u_i \\ v_i \end{pmatrix} \right) \quad m \times \left( -s_i, \begin{pmatrix} u_i \\ -v_i \end{pmatrix} \right) \quad (p - m) \times \left( 0, \begin{pmatrix} 0 \\ v_k \end{pmatrix} \right)$$

## Normalisation Subtlety

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} = \begin{pmatrix} U & U & 0 \\ V & -V & V_0 \end{pmatrix} \begin{pmatrix} S & 0 & 0 \\ 0 & -S & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} U^T & V^T \\ U^T & -V^T \\ 0 & V_0^T \end{pmatrix}$$

- Multiplying out we have

$$X = USV^T \quad X^T = VS^T U^T$$

- Now the vectors  $u_i$  and  $v_i$  form orthogonal sets as they satisfy

$$X^T X v = s^2 v \quad X X^T u = s^2 u$$

- But they are not normalised (since  $\begin{pmatrix} u_i \\ v_i \end{pmatrix}$  is normalised). If we define  $\tilde{U} = \sqrt{2}U$  and  $\tilde{V} = \sqrt{2}V$  we find

$$X = \tilde{U} \tilde{S} \tilde{V}^T \quad X^T = \tilde{V} \tilde{S}^T \tilde{U}^T$$

- Singular Value Decomposition
- General Linear Mappings
- Linear Regression Revisited

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

## Eigen Decomposition

- Consider an arbitrary  $m \times p$  matrix  $X$ , and construct the  $(m + p) \times (m + p)$  symmetric matrix,  $B$ ,

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

$\begin{pmatrix} u \\ v \end{pmatrix}$  is an eigenvector of  $B$  with eigenvalue  $s$

- We observe that

$$Xv = su \quad X^T u = sv$$

$$X^T X v = s X^T u = s^2 v \quad X X^T u = s X v = s^2 u$$

## Matrix Decomposition

- Stacking the eigenvectors into a matrix

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} = \begin{pmatrix} U & U & 0 \\ V & -V & V_0 \end{pmatrix} \begin{pmatrix} S & 0 & 0 \\ 0 & -S & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} U^T & V^T \\ U^T & -V^T \\ 0 & V_0^T \end{pmatrix}$$

- Since the vectors  $\begin{pmatrix} u_i \\ v_i \end{pmatrix}$  are eigenvectors of a symmetric matrix they form an orthogonal matrix if they are normalised.
- Multiply on the right by the transpose of the orthogonal matrix

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} = \begin{pmatrix} U & U & 0 \\ V & -V & V_0 \end{pmatrix} \begin{pmatrix} S & 0 & 0 \\ 0 & -S & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} U^T & V^T \\ U^T & -V^T \\ 0 & V_0^T \end{pmatrix}$$

## SVD

- Any matrix,  $X$ , can be written as  $X = USV^T$

- $U, V$  are orthogonal matrices
- $S = \text{diag}(s_1, s_2, \dots, s_r)$  with  $r = \min(m, p)$

- $s_i$  can always be chosen to be non-negative and are known as **singular values**

- If  $X$  is symmetric and positive semi-definite (e.g. a covariance matrix) then  $U = V$  and  $S = \Lambda$  (the matrix of eigenvalues)—SVD is the same as eigen-decomposition (for a general symmetric matrix  $s_i = |\lambda_i|$ )

- Singular value decomposition applies to both square and non-square matrices—they describe general linear mappings

## Spectrum of a Matrix

- The set of singular values (when ordered) is often referred to as the (singular value) spectrum of a matrix
- The largest singular value,  $s_{max}$ , provides a measure of the maximum change in the two norm of a vector

$$\mathbf{X}\mathbf{v}_{max} = \mathbf{U}\mathbf{S}\mathbf{V}^T\mathbf{v}_{max} = s_{max}\mathbf{u}_{max}$$

where  $\mathbf{u}_{max}$  and  $\mathbf{v}_{max}$  are the columns of the matrix  $\mathbf{U}$  and  $\mathbf{V}$  corresponding to  $s_{max}$  (note  $\|\mathbf{v}_i\| = \|\mathbf{u}_i\| = 1$ )

- $s_{max} = \|\mathbf{X}\|_2$  is known as the **spectral norm** or **2 norm** of a matrix satisfying

$$\|\mathbf{X}\mathbf{x}\|_2 \leq s_{max} \|\mathbf{x}\|_2$$

with equality when  $\mathbf{x} = \mathbf{v}_{max}$

## Economical Forms of SVD

- Often the rows or columns of the orthogonal matrices  $\mathbf{U}$  and  $\mathbf{V}$  that are not associated with a singular value are ignored

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{U} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \mathbf{0} \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{U} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \mathbf{0} \end{pmatrix}$$

- In numpy (which returns a vector  $\mathbf{s}$  and  $\mathbf{V}^T$ )

```
U, s, Vt = np.linalg.svd(X) # full: U is m x m
# economical: U is m x r
U, s, Vt = np.linalg.svd(X, full_matrices=False)
```

## General Matrix

- Recall that we can compute the SVD for any matrix,  $\mathbf{X}$
- As matrices describe the most general linear mapping

$$\mathbf{v} \rightarrow \mathcal{T}[\mathbf{v}] = \mathbf{X}\mathbf{v}$$

- We can use SVD to understand any linear mapping
- Thus any linear mapping can be seen as a rotation followed by a squashing or expansion independently in each coordinate followed by another rotation

## Determinants

- The determinant,  $|\mathbf{M}|$  of a matrix  $\mathbf{M}$  is defined for square matrices
- It describes the change in volume under the mapping
- Now for any two matrices  $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$
- Thus  $|\mathbf{M}| = |\mathbf{U}||\mathbf{S}||\mathbf{V}^T|$
- For an orthogonal matrix  $|\mathbf{U}| = \pm 1$  since  $\mathbf{U}\mathbf{U}^T = \mathbf{I} \Rightarrow |\mathbf{U}\mathbf{U}^T| = |\mathbf{I}| \Rightarrow |\mathbf{U}||\mathbf{U}^T| = 1$  or  $|\mathbf{U}|^2 = 1$
- Thus

$$|\mathbf{M}| = \pm |\mathbf{S}| = \pm \prod_i s_i$$

## Finding SVD

- Most libraries will compute the SVD for you
- In principle we can choose the smaller of the two matrices  $\mathbf{X}\mathbf{X}^T$  and  $\mathbf{X}^T\mathbf{X}$  and compute its eigenvalues (libraries avoid this: it squares the condition number)
- The singular values are the square root of the eigenvalues (notice that  $\mathbf{X}\mathbf{X}^T$  and  $\mathbf{X}^T\mathbf{X}$  are both positive semi-definite so the eigenvalues will be non-negative)
- We can then compute the  $\mathbf{U}$  matrix or  $\mathbf{V}$  matrix by multiplying through by  $\mathbf{X}$  or  $\mathbf{X}^T$  ( $\mathbf{U} = \mathbf{X}\mathbf{V}\mathbf{S}^{-1}$  and  $\mathbf{V} = \mathbf{X}^T\mathbf{U}\mathbf{S}^{-1}$ )
- In practice to perform PCA most people subtract the mean from their data and then perform SVD

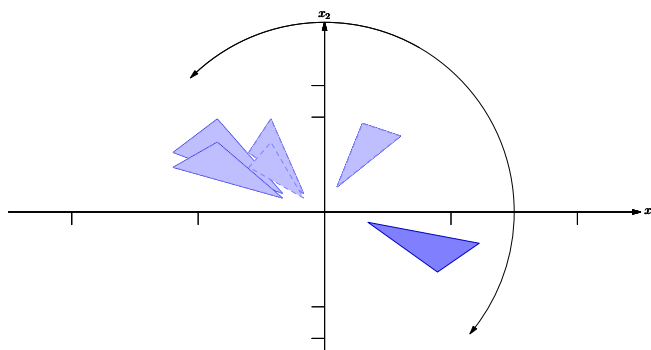
## Outline

- Singular Value Decomposition
- General Linear Mappings
- Linear Regression Revisited

$$\begin{pmatrix} \mathbf{0} & \mathbf{X} \\ \mathbf{X}^T & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} = \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} \mathbf{s}$$

## Matrices

$$\mathbf{M} = \begin{pmatrix} -0.45 & 1.9 \\ -0.77 & -0.025 \end{pmatrix} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = \begin{pmatrix} \cos(-175) & \sin(-175) \\ -\sin(-175) & \cos(-175) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75) & \sin(75) \\ -\sin(75) & \cos(75) \end{pmatrix}$$



## Non-Square Matrices

- When the matrices are non-square then the singular value matrix will either
  - Squash some directions to zero
  - Introduce new dimensions orthogonal to the vector

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{U} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \mathbf{0} \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{U} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{S} \\ \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{V}^T \\ \mathbf{0} \end{pmatrix}$$

- The rank of an arbitrary matrix is the number of non-zero singular values (also number of linearly independent rows or columns)

## Duality Revisited

- If  $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$  then

$$\begin{aligned}\mathbf{C} &= \mathbf{X}\mathbf{X}^T & \mathbf{D} &= \mathbf{X}^T\mathbf{X} \\ &= \mathbf{U}\mathbf{S}\mathbf{V}^T\mathbf{V}\mathbf{S}^T\mathbf{U}^T & &= \mathbf{V}\mathbf{S}^T\mathbf{U}^T\mathbf{U}\mathbf{S}\mathbf{V}^T \\ &= \mathbf{U}(\mathbf{S}\mathbf{S}^T)\mathbf{U}^T & &= \mathbf{V}(\mathbf{S}^T\mathbf{S})\mathbf{V}^T\end{aligned}$$

- If  $\mathbf{X}$  is an  $m \times p$  matrix then  $\mathbf{S}\mathbf{S}^T$  is an  $m \times m$  diagonal matrix with elements  $S_{ii}^2 = s_i^2$
- $\mathbf{S}^T\mathbf{S}$  is a  $p \times p$  matrix with elements  $S_{ii}^2 = s_i^2$
- $\mathbf{U}$  and  $\mathbf{V}$  are matrices of eigenvectors for  $\mathbf{C}$  and  $\mathbf{D}$
- The eigenvalues of  $\mathbf{X}\mathbf{X}^T$  and  $\mathbf{X}^T\mathbf{X}$  are  $\lambda_i = S_{ii}^2 = s_i^2$

## $\mathbf{S}\mathbf{S}^T$ and $\mathbf{S}^T\mathbf{S}$

$$\mathbf{S} = \begin{pmatrix} s_1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_m & 0 & 0 \dots & 0 \end{pmatrix}$$

$$\mathbf{S}^T\mathbf{S} = \begin{pmatrix} s_1^2 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2^2 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_m^2 & 0 & 0 \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & 0 \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 \dots & 0 \end{pmatrix}$$

$$\mathbf{S}\mathbf{S}^T = \begin{pmatrix} s_1^2 & 0 & \dots & 0 \\ 0 & s_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_m^2 \end{pmatrix}$$

## Having A Go

- It's really easy to verify this in Python

```
import numpy as np
X = np.random.rand(3, 2)
U, s, Vt = np.linalg.svd(X) # s: vector, Vt: V transposed
S = np.zeros_like(X)        # rebuild the 3 x 2 matrix S
np.fill_diagonal(S, s)
U @ S @ Vt                  # should equal X
U[:, 0] @ U[:, 1]           # columns of U are orthogonal
U.T @ U, U @ U.T            # both the identity
L, Va = np.linalg.eigh(X @ X.T) # eigenvalues, ascending
S @ S.T                      # the same values: s squared
```

- Test yourself!

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## Linear Regression

- Given a set of data  $\mathcal{D} = \{(\mathbf{x}_k, y_k) | k = 1, 2, \dots, m\}$
- In linear regression we try to fit a linear model

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{x}^T\mathbf{w}$$

- Which we fit by minimising the squared error loss

$$L(\mathbf{w}) = \sum_{k=1}^m (f(\mathbf{x}_k|\mathbf{w}) - y_k)^2$$

## Using SVD

- Using  $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$  then

$$\begin{aligned}\mathbf{X}^+ &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \\ &= (\mathbf{V}\mathbf{S}^T\mathbf{S}\mathbf{V}^T)^{-1}\mathbf{V}\mathbf{S}^T\mathbf{U}^T \\ &= \mathbf{V}(\mathbf{S}^T\mathbf{S})^{-1}\mathbf{V}^T\mathbf{V}\mathbf{S}^T\mathbf{U}^T \\ &= \mathbf{V}(\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T\mathbf{U}^T = \mathbf{V}\mathbf{S}^+\mathbf{U}^T\end{aligned}$$

- If  $m > p$

$$\mathbf{X}^T = \begin{pmatrix} | & | & | & | & | & | & | & | \\ s_1 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & s_3 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s_p & 0 & 0 \dots & 0 \end{pmatrix}, \mathbf{S}^T = \begin{pmatrix} s_1 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & s_3 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s_p & 0 & 0 \dots & 0 \end{pmatrix}$$

## Matrix Form

- In matrix form we write  $L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_m^T \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$$

- Then  $\nabla L(\mathbf{w}^*) = 0$  implies

$$\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{X}^+\mathbf{y}$$

- This is known as the pseudo-inverse

## Pseudo-Inverse of $\mathbf{S}$

$$\mathbf{S}^T\mathbf{S} = \begin{pmatrix} s_1^2 & 0 & \dots & 0 \\ 0 & s_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_p^2 \end{pmatrix}, (\mathbf{S}^T\mathbf{S})^{-1} = \begin{pmatrix} s_1^{-2} & 0 & \dots & 0 \\ 0 & s_2^{-2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_p^{-2} \end{pmatrix}$$

$$\mathbf{S}^+ = (\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T = \begin{pmatrix} s_1^{-1} & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2^{-1} & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & s_3^{-1} & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s_p^{-1} & 0 & 0 \dots & 0 \end{pmatrix}$$

## Moore–Penrose Pseudo-Inverse

- $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  needs every  $s_i > 0$ : it fails if  $m < p$  or the columns of  $\mathbf{X}$  are dependent
- The **Moore–Penrose** pseudo-inverse,  $\mathbf{X}^+ = \mathbf{V} \mathbf{S}^+ \mathbf{U}^\top$ , just skips the zero singular values:  $S_{ii}^+ = 1/s_i$  if  $s_i \neq 0$ , and 0 otherwise
- **Over-constrained** ( $m > p$ , all  $s_i > 0$ ): nothing changes,  $\mathbf{w} = \mathbf{X}^+ \mathbf{y}$  minimises  $\|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2$
- **Under-constrained** ( $m < p$ ): infinitely many  $\mathbf{w}$  fit the data, and  $\mathbf{X}^+ \mathbf{y}$  picks the one with the smallest  $\|\mathbf{w}\|$ 
  - ★ If  $s_i = 0$  then  $\mathbf{X} \mathbf{v}_i = \mathbf{0}$ : the data say nothing about  $\mathbf{w}$  along  $\mathbf{v}_i$ , so that component is set to zero
- In numpy: `np.linalg.pinv` or `np.linalg.lstsq`

## Regularisation

- Consider linear regression with a regulariser

$$\begin{aligned} \mathcal{L}(\mathbf{w}) &= \|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2 + \eta \|\mathbf{w}\|^2 \\ &= \mathbf{w}^\top (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I}) \mathbf{w} - 2 \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y} \end{aligned}$$

- Thus

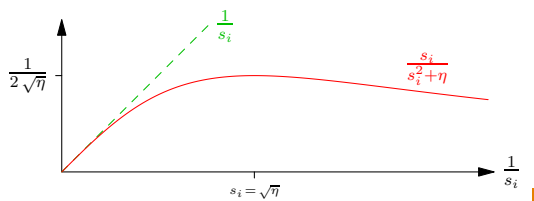
$$\nabla \mathcal{L}(\mathbf{w}) = 2(\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I}) \mathbf{w} - 2 \mathbf{X}^\top \mathbf{y}$$

- and  $\nabla \mathcal{L}(\mathbf{w}^*) = 0$  gives

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

## Effect of Regularisation

- Without regularisation if  $s_i = 0$  the problem would be ill-posed (even  $\mathbf{S}^+$  does not exist since  $s_i^{-1}$  would be ill defined) and if  $s_i$  is small then  $\mathbf{S}^+$  is ill-conditioned
- Using  $\hat{\mathbf{S}}^+ = (\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top$  instead of  $\mathbf{S}^+$  then



- Regularisation makes the machine much more stable (reduces the variance)

## III-Conditioned Data Matrix

- Recall that

$$\mathbf{w}^* = \mathbf{X}^+ \mathbf{y} = \mathbf{V} \mathbf{S}^+ \mathbf{U}^\top \mathbf{y}$$

- If any of the singular values of  $\mathbf{X}$  are small then  $\mathbf{S}^+$  will magnify components in that direction
- Any errors in the target  $\mathbf{y}$  will be magnified
- This leads to poor weights

## Regularisation Continued

- Using  $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^\top$

$$\begin{aligned} \mathbf{w}^* &= (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y} \\ &= \mathbf{V} (\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top \mathbf{U}^\top \mathbf{y} \end{aligned}$$

- where

$$(\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top = \begin{pmatrix} \frac{s_1}{s_1^2 + \eta} & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \frac{s_2}{s_2^2 + \eta} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 0 & \frac{s_3}{s_3^2 + \eta} & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{s_p}{s_p^2 + \eta} & 0 & 0 & \cdots & 0 \end{pmatrix}$$

## Summary

- Any matrix can be decomposed as  $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^\top$  where
  - ★  $\mathbf{U}$  and  $\mathbf{V}$  are orthogonal (rotation matrices)
  - ★  $\mathbf{S} = \text{diag}(s_1, \dots, s_r)$  is a diagonal matrix of non-negative singular values
- This describes the most general linear transform
- The transform exploits the duality between  $\mathbf{X} \mathbf{X}^\top$  and  $\mathbf{X}^\top \mathbf{X}$
- In linear regression the pseudo-inverse involves the reciprocal of the singular values, which can lead to poor generalisation
- Regularisation improves the conditioning of the “inverse” matrix